

SMART INDIA HACKATHON 2026

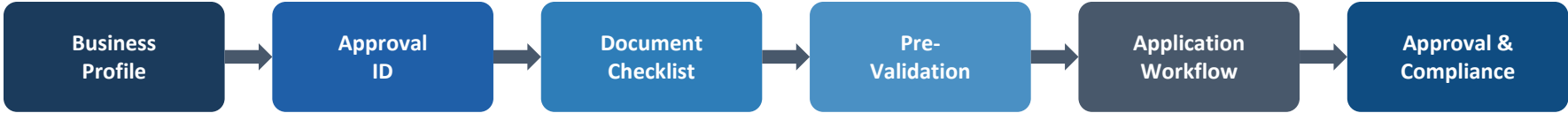


- Problem Statement ID – [SIH26130](#)
- Problem Statement Title- [Efficiency in streamlining industrial approvals, compliance processes, and access to government support services](#)
- Theme- [Miscellaneous](#)
- PS Category- [Software](#)
- Team ID- [SIH26130-XXX](#)
- Team Name- [Team Udyog Sarthi](#)
- College – ____ (fill before upload)
- Product – [Udyog Sarthi AI](#) – Intelligent Industrial Approval Platform



THE CHALLENGE

- Entrepreneurs face multiple portals & unclear approval requirements.
- No single source of truth for applicable licences, NOCs & schemes.
- Manual document checks cause delays & repeated rejections.
- Officers lack visibility into pending cases & bottlenecks across departments.



Our Solution (incl. Innovation)

- Unified platform for approvals, NOCs, inspections, renewals & schemes.
Profile → personalised roadmap (sector × location × size × stage) + checklist.
Pre-validation + parallel workflow + SLA tracking (entrepreneur & officer).
Cited TF-IDF assistant (no LLM) + renewal reminders (validity-table roadmap).
Pilot: 3 approvals × Sinnar MIDC, 20 users, 4 weeks; 68→31d, 41%→12%*.

Why Different — Innovation

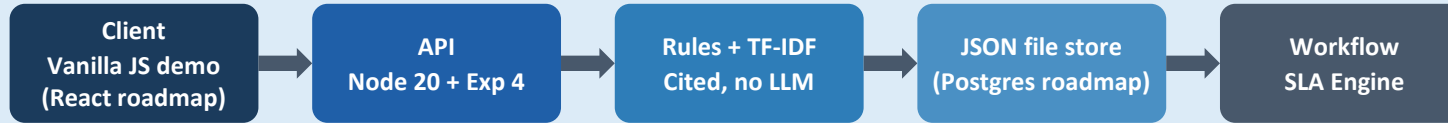
- TF-IDF retrieval with citations (P@1 0.75, no LLM), not generic listings.
Pre-submission checks + risk-based scrutiny.
Parallel workflows + SLA/bottleneck intelligence.

MAITRI | NSW | Udyog Sarthi

We complement, not duplicate: SSO + API over MAITRI/NSWS.

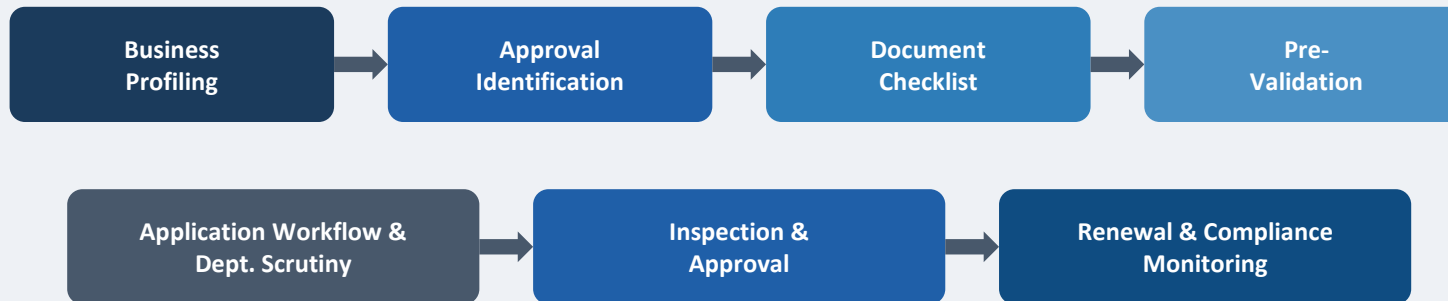
Capability	MAITRI	NSWS	Udyog Sarthi
Roadmap	Partial	Partial	Full
Pre-validation	No	No	Full
Parallel flow	No	No	Full
SLA/bottleneck	Basic	Basic	Full+pred.
Lifecycle	Partial	Partial	Full

System Architecture (Layered)



Demo now: Vanilla JS client → API (Node 20 + Express 4 REST) → Rules + TF-IDF retrieval v1 P@1 0.75 no LLM (cited) → JSON file store (pilot). Sidecars: demo auth/RBAC + validation + SLA sweeps. Phase-2: React, Postgres/pgvector, Tesseract OCR, SMS. NSW/MAITRI dashed.

Process Flowchart



Methodology: profile → ID → checklist → pre-val → workflow → scrutiny → inspect → approve → renew. Guardrail: cited answers only, human-in-loop; no legal advice without source.

Tech Stack Used

- Frontend: Vanilla JS offline demo (EN/MR toggle, React roadmap)

Backend: Node 20 + Express 4 REST + demo auth/RBAC

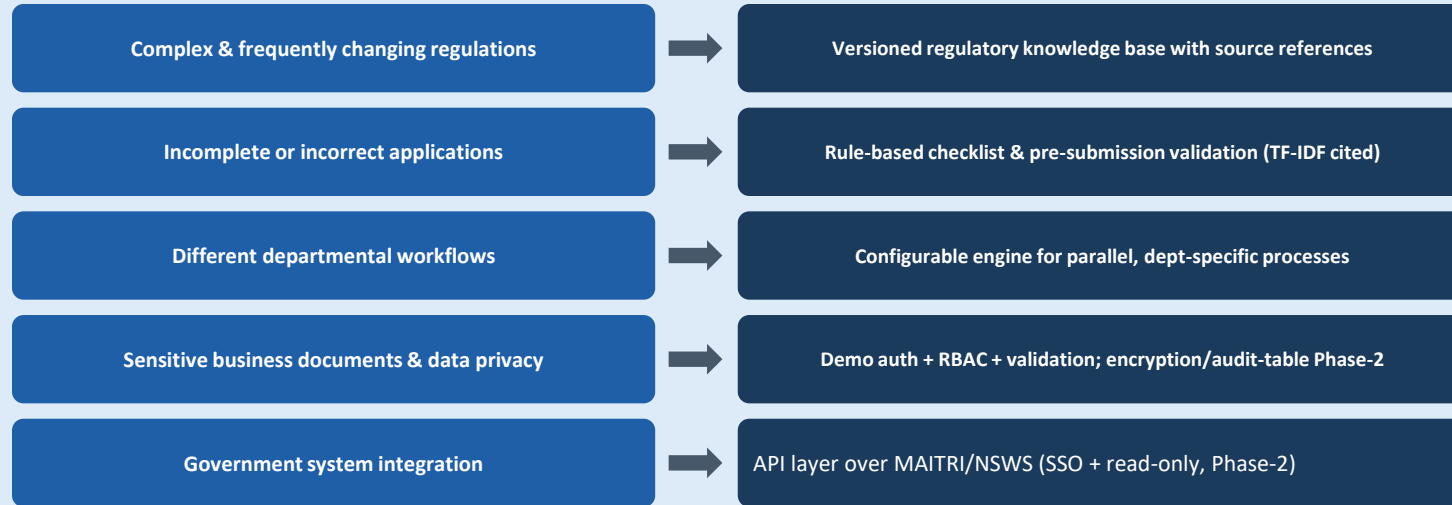
Data+AI: JSON file store (pilot) + TF-IDF rule+retrieval v1 P@1 0.75, no LLM, cited

Docs & Flow: filename+size+expiry validation now (Tesseract OCR Phase-2) + Rules + SLA sweeps

Ops: npm start :3000 offline; Postgres/SMS/audit-table/DPDP Phase-2

Eval: citation precision + checklist accuracy; human-in-loop.
NSWS/MAITRI connector dashed = Phase-2 (SSO + read-only).

Risk Assessment and Mitigation



Resource Requirements

- Laptop + npm start :3000 offline demo. JSON file store (pilot); Postgres Phase-2. TF-IDF no LLM (P@1 0.75); capped LLM Phase-2. Demo auth + validation + SLA sweeps; Tesseract/SMS/audit-table Phase-2.

Aligned With



Technical Feasibility

- Proven stack (Vanilla JS/Node/JSON) → rapid MVP; React/Postgres Phase-2.
- MVP: 3 approvals × Sinnar MIDC, 20 users, 4 weeks.
Modular: start small, expand dept-wise.

Implementation Approach

- Wk1-2 KB + auth; Wk3-4 rules + validation. Wk5 TF-IDF + dashboards; Wk6 pilot. Complement MAITRI/NSWS, not duplicate.

“ONE PLATFORM. EVERY APPROVAL. FULL TRANSPARENCY.”

STAKEHOLDER IMPACT

Entrepreneurs / MSMEs

- Personalized approval roadmap, fewer errors, reduced compliance effort, transparent tracking.

Government Departments

- Better application completeness, reduced repetitive scrutiny, improved workload visibility.

Government Officers

- Centralized case management, SLA monitoring, inspection scheduling, bottleneck identification.

Startups

- Faster understanding of registrations, licences, incentives & compliance obligations.

PILOT TARGETS & PROJECTED IMPACT*

- 68→31 days • 41%→12% resubmissions • ≥95% checklist • 100% SLA visibility (pilot depts)
- Pilot: 3 approvals (Udyam, Shops & Estt., Fire NOC) × Sinner MIDC, 20 users, 4 weeks. *Seed projection, pilot to validate. MR/HI/EN + CSC assist.

BROADER & LONG-TERM IMPACT

- **Government:** data-driven identification of approval delays & bottlenecks to improve ease of doing business.
- **Economic Impact:** reduced approval time & compliance cost can encourage entrepreneurship, investment & industrial growth.
- **Long-Term:** a scalable digital framework that can expand across sectors, districts & government services.

Government & Policy Resources

- MAITRI 2.0 — MIDC (04-Feb-2025, CM launch) — maitri.mahaonline.gov.in
Maharashtra Facilitation Act 2023 — time-bound disposal + deemed provision

National Frameworks

- NSWS — 325 Central + ~2200 State approvals — nsws.gov.in
India Code — indiacode.nic.in; MCA21, GSTN (read-only, Phase-2)

Industry & Facilitation

- Udyam Registration — udyamregistration.gov.in; PSI-2019 incentives
Fire NOC / Shops & Estt. — Sinnar MIDC pilot (3 approvals)

Digital Governance Reports

- EoDB / NSWS dashboards — baseline 68d, 41% resubmissions*
Pilot target 31d, 12% — *seed projection, pilot to validate